

Geophysics III

Steady-State Geotherms

This problem set is worth a total of 20 points to the total problem set score.

1. [4 pts.] Thermal conductivity

Using the mineralogical composition chart and the thermal conductivities listed in the table below, compute the thermal conductivity for

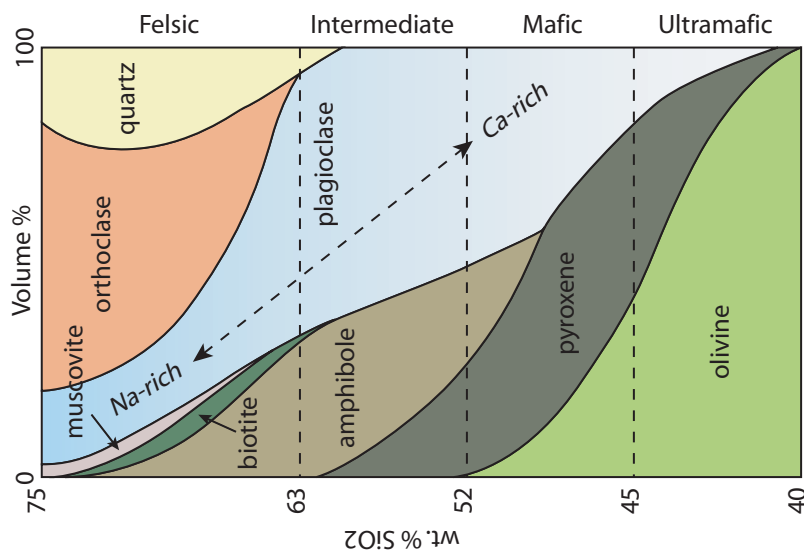


Figure 1: Mineralogical compositions of igneous rocks.

- (a) granite
- (b) diorite
- (c) gabbro, assuming all pyroxene is clinopyroxene.
- (d) peridotite

Phase changes associated with pressure turn plagioclase into spinel and then into garnet. Therefore, compute the thermal conductivity of peridotite with a composition of 67% olivine, 17% orthopyroxene, 8% clinopyroxene, and 8% garnet.

Table 1: Thermal conductivity of common materials.

mineral	k $\text{W m}^{-1} \text{K}^{-1}$
α -quartz	8.8
orthoclase	1.8
plagioclase, felsic	1.8
plagioclase, mafic	1.6
mica, muscovite	2.1
mica, biotite	1.9
hornblende (amphibole)	2.7
clinopyroxene	4.3
orthopyroxene	3.4
olivine	4.8
spinel	9
garnet	3.9

All conductivities estimated at 25 °C.

2. [10 pts.] Continental Geotherms

This problem is completed with the python-based codes, `ss_isostasy.py`, which you can find on the course GitHub page. Run in VSCode or python.

On paper:

- (a) Let's start by computing a geotherm (temperature as a function of depth) roughly consistent with a craton. We will keep it relatively simple by dividing the lithosphere into three layers: an upper crust; lower crust; and mantle lithosphere (Figure 2).

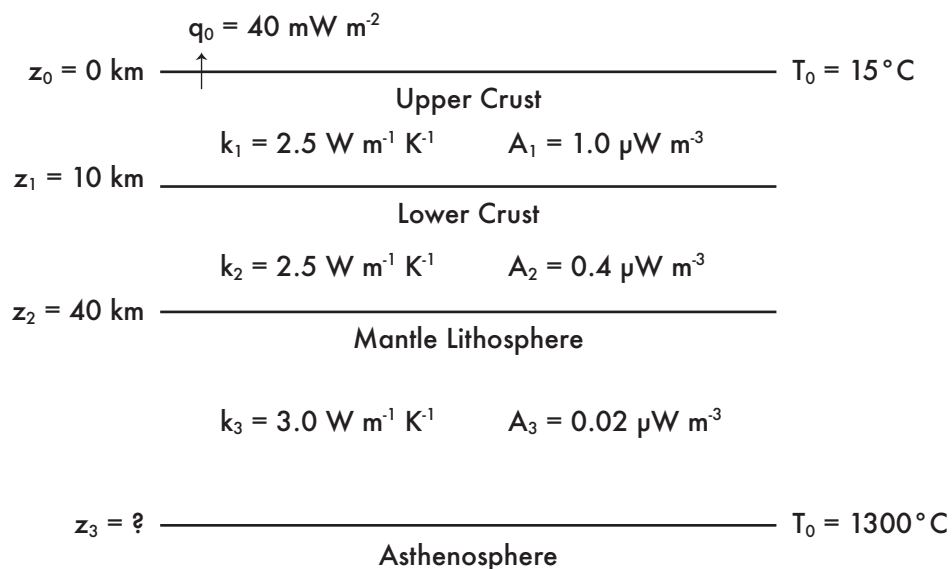


Figure 2: A three layered geotherm model.

Using the geotherm solution in Appendix A, compute the temperature at each of the interfaces above (except the lithosphere–asthenosphere boundary). Use $q_0 = 40 \text{ mW m}^{-2}$ and $A_1 = 1.0 \mu\text{W m}^{-3}$. For the lithosphere–asthenosphere boundary, assume the con-

vecting mantle temperature is 1300°C. This geotherm will be our reference for all future elevation computations.

Plot your result and interpolate between layers. **What should the shape of your curve be?**

On computer:

Complete the rest using `ss_isostasy.py`, a python calculator that will compute geotherms,

- (a) Now, computing the reference geotherm roughly consistent with a craton, using the same layering as above (Figure 2). Entering the appropriate properties into the table and text boxes.
- (b) What causes the kinks in the curve, especially at the Moho?
- (c) Now let's examine the sensitivity of geotherms to variations in heat production. This test will illustrate why reducing uncertainties in crustal heat production are crucial for properly estimating lithospheric temperatures.

Compute three geotherms: (1) increase upper crustal heat production by 25%; (2) increase lower crustal heat production by 25%; and (3) increase mantle lithospheric heat production by 25%. All other properties remain the same as those in Figure 2.

Does increasing heat production increase or decrease the lithospheric temperatures? How much does each test affect the temperature at the Moho? How about the depth at which the adiabat is intersected?

- (d) Now let's compute a family of geotherms with heat flow values of 40, 60, 80, 100 mW m⁻². The upper crustal heat production and surface heat flow are the only value we will vary. To determine the heat production we will assume a partition coefficient of $P = 0.75$. The upper crustal heat production can be computed by

$$A_1 = (1 - P) \frac{q_0}{D}, \quad (1)$$

where D is the thickness of the upper crust, 10 km.

Using your combinations of q_0 and A_1 above, compute each geotherm. Aside from q_0 and A_1 , all other inputs to the model should be the same.

- (e) Compute the elevations of each geotherm from the geotherm family above (the equations it uses are found in Appendix B). Use the 40 mW m⁻² geotherm as the reference. Assume the thermal expansivity, α , is $3 \times 10^{-5} \text{ K}^{-1}$. Hint: you won't need to compute the elevation for the 40 mW m⁻² geotherm since the reference elevation is defined as 0 km. Plot the elevations as a function of surface heat flow. To compute the total elevation, you will need to compute the individual elevation contributions for four steps, one for each lithospheric layer and one for depths below the intersection with the mantle adiabat.
How much is the elevation affected by heat flow over the range of typical continental heat flows (40 to 100 mW m⁻²)?

- (f) Now let's see if we can estimate the heat production of the Australian provinces. Start by computing another set of geotherms with heat flow $q_0 = 80 \text{ mW m}^{-2}$ with $P = (0.75, 0.55, 0.45, 0.35)$.

Repeat the procedure from 3 for each new P . Plot the new geotherm family.

- (g) Compute the elevation differences for geotherms with $q_0 = 80 \text{ mW m}^{-2}$. Use the $q_0 = 40 \text{ mW m}^{-2}$ geotherm as the reference.

Repeat the steps used to produce the geotherm family and elevations above with the new heat production parameters. Plot each result.

Which value of heat production is most consistent with the reference cratonic geotherm? Could this be the explanation for our elevation anomalies?

3. [6 pts.] Thermal stability of cratons

Cratons are typically viewed as regions that are thermally cool, which contributes to their longevity/stability. In this problem, you will explore the effect of crustal radioactivity on crustal temperatures both for the present-day Earth and at a time 3 Ga when many of these stable regions were being first created.

This problem is accompanied by a spreadsheet (see template online) that you will need to complete as part of the exercises below.

- Compute the average heat production for a granitic and gabbroic compositions at the present day for crystallization ages of <200 Ma, from 2000 to 2800 Ma, and prior to 2800 Ma. The formulae should be directly entered into the worksheet labeled "heat production" and copied and modified as necessary for the other two rock ages.
Note: the concentration of potassium is provided as K_2O , so you will need to convert to K. And both K and U have multiple isotopes, so you will need to account for that in your calculation using the constants provided in the worksheet.
- In the same sheet, compute the average heat production for these same rock types and ages adjusted to a time 3 billion years ago (3 Ga).
- The values you previously computed will be automatically copied into the worksheet labeled "temperature". We want to compute a set of geotherms in this new sheet for a simplified crust created for six crustal heat production scenarios corresponding to the different age rocks at both present day and at 3 Ga.

We will make some simplifying assumptions for a typically moderate crustal heat flow:

- upper crust is 16 km thick;
- lower crust is 24 km thick;
- basal heat flow is 36 mW m^{-2} —use the conservation of energy to compute the heat flow at the top of each layer, all the way back to the surface; and
- thermal conductivity is temperature variable—the equations for thermal conductivity are already computed for you and will automatically update once you have computed the temperature.

If you enter the equations correctly into the first empty heat flow and temperature cells, you should be able to copy them to the end of each column. With the appropriate \$ on cell columns/rows, you will also be able to copy the heat flow and temperature columns from the first scenario to all others.

- The remainder of the problem leads up to computing heat production for three hypothetical regions at a point 3 Ga in the past so that the geotherms in the past. Start by determining the relationship between half-life and radioactive decay constants. The concentration of a radioelement, $P(t)$, is given by

$$P(t) = P_0 \exp(-\lambda t), \quad (2)$$

where P_0 is the initial concentration, and λ is the decay constant. Hint: a half-life is defined as the time at which the ratio of the radioelement concentration is one-half the initial concentration. Show your work.

- Compute the heat production of for the heat production concentrations at a point in time 3 Ga in the past. To compute the heat production in the past, you will need to adjust the concentrations of the radioisotopes.
- Compute one-dimensional, steady-state geotherms for each of the situations above to the Moho at a time 3 Ga in the past.
- Explain, in light of your computations, how changes in heat production concentrations

of igneous rocks helps lead to stability of Archean lithosphere.

A Computing geotherms

The solution for the temperature at the base of the first layer, T_1 , can then be estimated using the homogeneous solution,

$$T_1 = T_0 + \frac{q_0}{k_1}(z_1 - z_0) - \frac{A_1}{2k_1}(z_1 - z_0)^2. \quad (3)$$

The heat flow at the base of the first layer is equal to

$$q_1 = q_0 - A_1(z_1 - z_0). \quad (4)$$

The homogeneous solution can then be used to estimate the temperature at the base of the second layer using, T_1 , as the temperature at the top and q_1 as the heat flow. The temperature of all subsequent layers can be found using the same method boot-strapping down through the domain of interest.

Solving for the temperature at the base of the i -th layer,

$$T_{i+1} = T_i + \frac{q_i}{k_{i+1}}(z_{i+1} - z_i) - \frac{A_{i+1}}{2k_{i+1}}(z_{i+1} - z_i)^2. \quad (5)$$

It is also necessary to know the heat loss at the base of each layer in order to compute temperature. Since heat flow is determined by the heat flowing into a layer plus the heat generated within, the heat flow at the bottom of each layer is given by

$$q_{i+1} = q_i - A_{i+1}(z_{i+1} - z_i). \quad (6)$$

This solution forms the basis of most 1-D steady-state geotherms.

B Thermal isostasy

It can be shown that the relative elevation difference between a region with geotherm $T(z)$ and a reference region T_{ref} is given by

$$\varepsilon = \alpha \int_0^{z_{\text{max}}} [T(z) - T_{\text{ref}}] dz. \quad (7)$$

For our piecewise geotherms, we can compute an elevation within each depth interval by

$$\varepsilon_i = \alpha \int_{z_i}^{z_{i+1}} [T(z) - T_{\text{ref}}] dz, \quad (8)$$

and summing the individual steps

$$\varepsilon = \sum_{i=1}^N \varepsilon_i. \quad (9)$$

Within each step, we can integrate difference in temperature by integrating each geotherm separately and taking the difference since the integral operator is linear,

$$\varepsilon_i = \alpha \left[\int_{z_i}^{z_{i+1}} T(z) dz - \int_{z_i}^{z_{i+1}} T_{\text{ref}} dz \right]. \quad (10)$$

The integral of a single geotherm is given by

$$\int_{z_i}^{z_{i+1}} T(z) dz = \int_{z_i}^{z_{i+1}} \left[T_i + \frac{q_i}{k_{i+1}}(z - z_i) - \frac{A_{i+1}}{2k_{i+1}}(z - z_i)^2 \right] dz, \quad (11)$$

$$= T_i z + \frac{q_i}{2k_{i+1}}(z - z_i)^2 - \frac{A_{i+1}}{6k_{i+1}}(z - z_i)^3 \Big|_{z_i}^{z_{i+1}}, \quad (12)$$

$$= T_i(z_{i+1} - z_i) + \frac{q_i}{2k_{i+1}}(z_{i+1} - z_i)^2 - \frac{A_{i+1}}{6k_{i+1}}(z_{i+1} - z_i)^3. \quad (13)$$